

Introduction

Financial markets are driven by large volumes of unstructured news containing complex and evolving relationships. Over four semesters this project built the machinery to extract that structure; what remained was the gap between a pipeline the team runs and a system a sponsor uses.

This semester closes it. We integrate every prior team's work: Fall 2025's extraction pipeline, Summer 2025's taxonomy and Judge LLM, Spring 2025's canonicalization and GAT vs GraphSAGE GNN encoder comparisons, and Spring 2026's leakage fix and NOTEARS causal features. On top we add a three-LLM refinement chain (extract, judge, refine) with corpus-wide canonicalization, and run it across four corpora: the full 15-year FNSPID archive on a local model at zero API cost, a GPT-5.5 refined corpus for a controlled quality experiment, and two live corpora (STOXX Europe 600, S&P 100) retrained without any human in the loop. A VM-to-UI publishing pipeline ships everything as a public web application with graph, prediction, and factor views, an MCP connector, and a documented handover. It is ready to use.

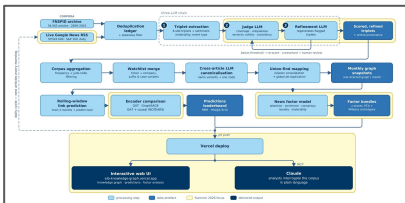


Figure 1: Unified End-to-End Pipeline

Multi-LLM Benchmark & Model Selection

The team evaluated the performance of 7 LLM models on the triplet extraction part using 4 metrics: uniqueness, coverage, semantic validity and consistency:

	Claude Sonnet 5	GPT 5.5	Gemini	Qwen 2.5b	Mistral	Llama 3.1	Kimi K3
Coverage	0.68	0.72	0.74	0.69	0.69	0.66	0.67
Uniqueness	0.89	0.88	0.83	0.92	0.89	0.88	0.87
Semantic Validity	0.83	0.85	0.84	0.81	0.80	0.73	0.86
Consistency	0.94	0.95	0.92	0.94	0.91	0.90	0.94

Table 1: LLM Benchmark Comparison

No model dominates across all four metrics: GPT-5.5 leads on semantic validity (0.85) and ties for best consistency (0.95), while Qwen 2.5b posts the highest uniqueness (0.92) and ties on consistency. The other five models cluster in the middle without topping any metric, making them less differentiated for the pipeline. GPT-5.5 and Qwen 2.5b each maximize a different axis of extraction quality (reliability/validity vs. distinctness), which is why we chose to keep them for their respective advantages. We ran two versions of the FNSPID corpus: one with GPT 5.5 extraction (with API costs) from 2019-2020, and a full corpus run with 0 API costs with Qwen 2.5b.

Refinements

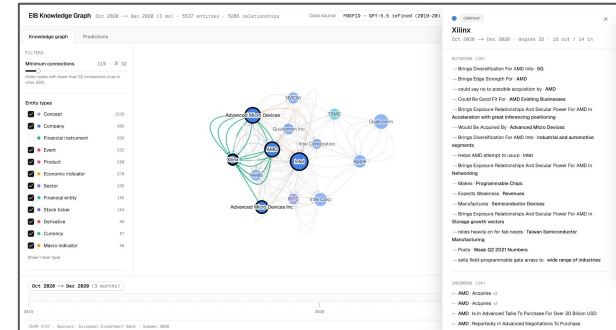


Figure 2: Knowledge Graph UI

Prediction Models

Predictions use a two part architecture for temporal link prediction in the style of FinDKG: a GNN encoder turns each entity into an embedding, and a shared dot product prediction head scores candidate pairs, trained with a cross entropy ranking loss on a rolling three month window. The encoder is swappable (GAT, GraphSAGE, GAT with NOTEARS causal features); the head and training recipe never change, so metric differences isolate the encoder. Served as monthly leaderboards with novelty z-scores and predicted co-movers, with a Model encoder toggle and per-encoder metrics table in the UI.

Encoder comparison on the GPT-5.5 corpus (23 backtest months):

- GAT with edge features (relation, category, weight): MRR 0.863
- GAT without edge features: MRR 0.874
- GraphSAGE (mean aggregation, revisiting Spring 2025): MRR 0.942
- GAT + causal features (NOTEARS, ported from Spring 2026): MRR 0.864

Live Corpora & the News Factor Model

Everything upstream ran on a static research dataset (FNSPID's news ends 9 Jan 2024). We therefore stood up **two fully live corpora**, STOXX Europe 600 (sponsor-relevant, six sectors) and the full S&P 100 (~101 constituents), and these are not a side demo: they exercise the entire deployed pipeline, end to end, every day. A 07:30 UTC cron on the team VM fetches per-ticker news, extracts with an enriched schema (each triplet carries sentiment, dollar materiality, and an event type), passes the same Judge-and-refinement chain and three-pass canonicalization as the historical corpus, and merges the result into monthly knowledge-graph snapshots — so the interactive graph, entity filters, and industry views in the UI grow with the news, with no human in the loop and zero API spend (using Qwen 2.5b via Ollama). Once a corpus accumulates enough months, the cron's GAT stage activates automatically: the CE No-Edge-Feats configuration retrains overall accumulated months and republishes the predictions leaderboard each morning. STOXX already predicts live and the S&P tab flips on the first morning its history supports a training window.

On top of the enriched triplets sits the Fama-French-inspired news factor model in a separate tab: every entity scored daily on Attention, Sentiment, Consensus, Novelty, and Materiality (based on Summer 2025's Judge-LLM design); z-scored factors feed KMeans clustering in the full factor space, discovering news archetypes with no sector labels supplied (e.g., Novo Nordisk / Shell / Volkswagen emerge as the heavily-covered-but-contested cluster), with PCA projecting to two axes for display only. The live corpora are thus the proof of the semester's thesis in miniature: the same extraction, quality control, graph construction, and prediction machinery validated on fifteen years of history, running itself in production on today's news.

Deployment

The system ships as a finished product: <https://eib-knowledge-graph.vercel.app>.

The frontend is Next.js + TypeScript with Tailwind/shaden styling and a custom d3-force graph engine, chosen so the entire app compiles to a static site. All data is plain JSON on a CDN, so there is no server, no database, and nothing to maintain or pay for. A daily cron on the team VM does the only moving work: scrape, three-LLM chain, canonicalize, retrain. Git push redeploys the site automatically; a pulsing Live badge shows the last real pipeline run.

One dropdown switches between news corpora (15-year Cull Corpus FNSPID (Qwen 2.5), 2019-2021 FNSPID (GPT 5.5), live STOXX 600, live S&P 100). Per corpus, separate tabs show the knowledge graph, the prediction models, and the factor-model archetypes for the two live data sources, all navigated by draggable timelines. Analysts can also connect Claude to the corpus through a built-in MCP endpoint (public URL, no key) and interrogate the news behind any node in plain language.

Inheritance is designed in: a two-tier handover runbook (maintain vs. full rerun) is linked from the app's own settings; production needs no API keys anywhere, and every account binding (repo, hosting, VM) is swappable. Unmaintained, the site degrades gracefully into a permanent static archive; maintained, it costs one cron on one GPU. It is ready for the sponsor to use today.

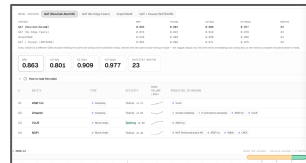


Figure 3: Predictions UI



Figure 4: Factor analysis UI

Final Thoughts

The semester closes with the loop fully shut. Five semesters of machinery, extraction, judging, refinement, canonicalization, and link prediction, now operate as one deployed system rather than a set of experiments. It has processed fifteen years of financial news end to end, proven that data quality drives predictive performance more than model choice, and benchmarked every encoder the lineage proposed on a single shared architecture. It sustains itself: two live corpora retrain every morning with **no human in the loop**, at zero API cost on a benchmark-validated local model. And it is built to outlive us: public, documented, keyless in production, with a handover for whoever comes next. What began as a triplet-extraction experiment is now a sponsor-ready system. It is ready to use.

References

- [1] Li, X. (2023). FinDKG: Dynamic Knowledge Graph with Large Language Models for Global Finance. SSRN 4608445.
- [2] Dong, Z. et al. (2024). FNSPID: A Comprehensive Financial News Dataset in Time Series. arXiv:2402.06698.
- [3] Lee, G., Gao, J., Lu, X., Cuadros, A. (2025, Dec). LLM and Graphs for Financial Texts.
- [4] Chen, X., Chen, X., Yang, Y. (2026, May). LLM and Graphs for Financial Texts.